Integration Test Scenarios

Simulation + Field Tests

8-Robot Heterogeneous Swarm — 60 Scenarios

문서번호 SAN-TST-INT-001

v1.5

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Version	v1.5
Classification	Test Specification
Preceding Docs	SAN-SDD-SWARM-001 v1.5, SAN-IDS-CMD-001 v1.5, SAN-SDD-SUR-001 v1.5
Test Environment	Gazebo Harmonic + ROS 2 Humble (Sim) + Real robot (Field)
Scenario Count	66 (Sim 59 + Real 7) — +6 new S18 scenarios in v1.5
KPP Mapping	10 KPPs directly measured

Revision History

Version	Changes
v1.0	Initial release. 42 scenarios (S1-S14).
v1.1	Added 6 new S15 scenarios (total 48): 360° surveillance, sweep detection, Hub dual SBC, SLAM, GStreamer SRT, multi-video.
v1.2	PM decisions D1~D7 + A1~A5 applied (total 54 scenarios): (D1) 9 -> 8 robots (S15 scenarios adjusted for robot count); (A2) **6 new S16 scenarios** — DEMO 6-phase sequencer verification; (A5) Section 2.5 **bench test environment** added — lightweight CI without Gazebo; Regression CI matrix gains bench tier (PR-level 10 min).
v1.3	Cost Map + LTE Redundancy + SLAM 5s testing added (60 scenarios total): (1) 6 new S17 scenarios — E1 ground filter, 235mm obstacle, 220mm ditch, 30 deg slope, Cost Map KPP, LTE fail-over; (2) S15-4 SLAM consistency — period 30s/60s -> 5s updated; (3) bench env adds Cost Map dummy simulation.
v1.4	Deputy UGV + Limp Mode tests added (66 scenarios total): (1) **6 new S18 scenarios** — Leader->Deputy succession, Leader+Deputy->Hub succession, Hub->Deputy takeover, 3-fail->battery follower, Limp Mode entry/exit; (2) Section 1.2 squadron table updated (Deputy UGV S3 + Followers 1~5).
v1.5	**DCN-2026-001 applied (2026-05-12): (1) S18-1~6 scenario bodies formally added (§11, was Revision-History-only in v1.4); (2) §1.2 Total 60 → 66 (Sim 59 + Real 7); (3) §2.3 9-robot test → 8-robot Standard + 4-robot Min variants; (4) §11 Scenario Index renumbered to §12 with header "66 total"; (5) Related Docs version pins updated v1.4 → v1.5; (6) Body references "9-robot V" / "9 robots" → "8-robot" throughout.**

Table of Contents

1. Overview

1.1 Document Purpose

This document defines integration test scenarios for the SAN swarm-controlled small tactical robot system. Each scenario provides a standard procedure to verify KPP satisfaction and regression stability under simulation (Gazebo Harmonic) or real-robot environment.

1.2 Scenario Classification

Category	Count	Environment	Purpose
Basic function	15	Gazebo	Verify core functions
KPP verification	12	Gazebo + real robot	Quantitative measurement of 10 KPPs
DEMO sequence	6	Gazebo + real robot	DEMO 6-phase auto-demo verification (v1.2)
Cost Map + LTE	6	Gazebo + bench + real	Cost Map avoidance + LTE redundancy (v1.3 new)
Stress	9	Gazebo	Comm loss, jamming, multi-obstacle extremes
Regression	12	Gazebo CI	Auto-verify retained functions
Total	60	—	—

1.3 KPP Mapping

KPP	Target	Verification Scenarios	Measurement
Formation hold avg error	≤ 2 m	S2-2, S6-2, S12-1, S14-1	Follower actual vs desired RMSE
Proximity reaction	≤ 300 ms	S9-1, S9-2, S12-2	Obstacle detect → avoidance start
Swarm control latency	≤ 150 ms	S7-3, S8-2, S12-3	Message timestamp differences
Leader reform	≤ 10 sec	S10-3, S12-4, S14-2	Leader OFF → new Leader stabilize

KPP	Target	Verification Scenarios	Measurement
Assembly success rate	≥ 95%	S4-1, S4-2, S12-5, S14-3	100 reps success count
Drone detection (RF)	≤ 2 km	S13-1, S13-2	RF scanner range
Real-time video	≤ 200 ms	S15-5	GStreamer SRT latency
Target classification	≥ 90% (≤ 2s)	S15-2	Pan-tilt sweep detection rate
Pan-tilt tracking	≤ 0.05° (500m, 3m/s)	S13-3	Track mode precision
Firing accuracy	≥ 90% (150m, 45cm)	Live-fire range (separate)	RCWS precision strike

2. Test Environments

2.1 Simulation Environment

Item	Specification
OS	Ubuntu 22.04 LTS
ROS 2	Humble Hawksbill
Simulator	Gazebo Harmonic (Sim 8)
UGV model	330kg tracked (custom URDF + Gazebo plugin)
Leader model	Unitree Go2 URDF (champ_description)
LiDAR model	Robosense E1 (UGV), Unitree L1 (Go2) — v1.1 update
Comm sim	ros2_network_sim (latency/loss injection), OpenWrt mesh sim
RTK GNSS	gnss_sim with RTK fixed/float toggle
Behavior Tree	BehaviorTree.CPP v4 + Groot2
Test runner	pytest + ros2 launch_testing
CI/CD	GitHub Actions + Docker (regression suite)

2.2 Simulation Worlds (5 types)

World ID	Name	Use	Key Features
W1	open_field	Open-area recon	50x50 m flat, scattered obstacles
W2	forest	Wooded area	100-300 procedurally generated trees
W3	urban_blocks	Urban alleys	5x5 blocks, corridor width 2-4m
W4	mountain_dem	Mountain/valley	DEM heightmap, cliff edges
W5	mixed_combat	Combined ops	Open + woods + narrow + cliff

2.3 Real-Robot Environment

Item	Specification
Location	Internal test site (Phase 1) + integrated training ground (Phase 2)
3-robot test	Unitree Go2 1 + tracked UGV 2 (Robosense E1)
8-robot test (Standard) + 4-robot test (Minimum)	Unitree Go2 1 + Hub UGV 1 (dual SBC) + tracked Follower UGV 6 (Robosense E1)
RTK base	Embedded in operator tablet or base-station deployment
Wi-Fi 6 mesh	OpenWrt 24.10, 4-6 APs distributed
Weather	Clear + rain/fog partial conditions
Live-fire test	Certified range separately, live ammo

2.4 Measurement Tools

- rosbag2 — record all messages
- PlotJuggler — real-time graph analysis
- Foxglove Studio — multi-message synchronized visualization
- ros2_kpp_recorder — automatic KPP measurement (in-house)
- Ground truth: Gazebo /world/model or real-robot external RTK
- GStreamer analysis: ffprobe, GStreamer GST_DEBUG=*:5
- Mesh diagnostics: batctl, mwan3 status, openwrt-stats

2.5 bench Test Environment — Lightweight CI (v1.2 new)

Environment for fast regression on internal PC without Gazebo. Runs automatically on every PR with 10-min results.

2.5.1 Environment

Item	Specification
Environment	Ubuntu 22.04 + ROS 2 Humble (Docker container)
Hardware	x86_64 or ARM64 PC, 8GB RAM, no GPU required
Dummy node	combat_robot_test_dummy_node — IMU/Pan-Tilt/Detector/Gun sim publish
Target deployment_mode	bench (yaml overlay, HW watchdog OFF, dummy data ON)

Item	Specification
Verification scope	FSM transitions, message communication, swarm relay, deputy fail-over, S16 DEMO sequence
Not possible	Real-time video, AI inference (Hailo8), SLAM consistency (Gazebo needed)

2.5.2 CI Tier Hierarchy

Tier	Frequency	Environment	Time	Pass criteria
L1 bench	Every PR	Docker + dummy	10 min	S1, S3, S4, S10, S15-1, S16 PASS
L2 Gazebo	Nightly	Gazebo Harmonic	1-2 hrs	S5-S11, S15-2~6 PASS
L3 Real (HIL)	Weekly	Real robot 3 units	2 hrs	S13 PASS
L4 Real full 8	Monthly/release	Real robot full set	4 hrs	S14, S15-5 PASS

3. Test Scenarios — Sprint 1-4 (Tracking + Formation)

3.1 S1 — Body-Frame Offset PID Tracking

S1-1 — Stationary Leader Follower Placement

Attribute	Value
Purpose	Verify follower reaches desired offset when Leader stationary
Environment	W1 open_field
Setup	Leader (0,0), Follower (10,5) start, target offset (-5,0)
Steps	(1) Leader pose publish → (2) PID tracker enable → (3) 30s observation
Pass	Within 30s reach (-5,0) ±0.5m, RMS oscillation <0.2m, overshoot <10%

S1-2 — Linear Motion Leader Tracking (3-robot V)

Attribute	Value
Purpose	Verify V-formation (d=5m, θ=90°) maintained during Leader linear motion
Environment	W1, 50m straight
Setup	Leader 1.0 m/s, 2 followers V-formation
Pass	50m run, avg formation error ≤ 2.0m, max ≤ 3.0m

S1-3 — Curved Path Tracking (3-robot V)

Attribute	Value
Purpose	Verify V hold on curved paths (radius 5m, 10m, 20m)
Environment	W1, circular path
Pass	Avg error ≤ 2.5m at each radius (5% KPP margin)

3.2 S2 — Hybrid A* Global Path Planning

S2-1 — Static Obstacle Avoidance Path

Attribute	Value
Environment	W2 forest (50x50m, 100 trees)
Setup	Leader (0,0), goal (40,40), static OccupancyGrid
Pass	Hybrid A* path generated in 5s, 0 collisions, path/straight ratio ≤1.5x

S2-2 — Dynamic Replan (Obstacle Added)

Attribute	Value
Environment	W2, new obstacle spawn during driving
Pass	Replan ≤ 2s, collision avoided, formation hold avg ≤ 2m

3.3 S3 — 1-Second Prediction Broadcast

S3-1 — FollowerTargetMessage 100ms Period

Attribute	Value
Environment	W1, Wi-Fi6 mesh sim (50ms latency + 1% loss)
Setup	Leader 1.0 m/s, 3 followers, /swarm/predict/follower_target rosbag
Pass	10Hz ±10% publish, deadline < 200ms, sequence_num monotonic

3.4 S4 — Hungarian Slot Assignment

S4-1 — V → Column Transition (Assembly Success Rate)

Attribute	Value
Environment	W1, 3 robots V start
Setup	"Formation transition: column" command 100 repetitions
Pass	Assembly success ≥ 95% (KPP), avg transition ≤ 10s

S4-2 — Column → Diamond Transition

Attribute	Value
Setup	4-robot column → diamond, 100 reps
Pass	Success ≥ 95%, 0 collisions, avg transition ≤ 8s

4. Test Scenarios — Sprint 5-8 (Network + Robustness)

4.1 S5 — All 9 Formation Types

S5-1 — All 9 Formations Entry/Stabilization

Attribute	Value
Environment	W1, 8-robot squadron
Setup	Each formation 5 entries (45 attempts)
Pass	Overall success ≥ 95%, 0 collisions per formation

S5-2 — Formation Auto-Transition (Terrain-Aware)

Attribute	Value
Environment	W4 mountain (cliff edges)
Pass	Auto column entry within 1m of cliff, 5s override window functional

4.2 S6 — V Formation + 8-Robot Stability

S6-1 — 8-Robot V Linear Drive (5 min)

Attribute	Value
Environment	W1, 8-robot V ($d=5m$, $\theta=60°$)
Pass	5-min run, avg error ≤ 2m, max error ≤ 4m

S6-2 — 8-Robot V Curved Drive

Attribute	Value
Setup	10m radius circular path, 8-robot V
Pass	Formation hold KPP ≤ 2m met

4.3 S7 — Wi-Fi 6 Mesh + DDS Communication

S7-1 — Mesh Node Add/Remove Hop

Attribute	Value
Environment	OpenWrt mesh sim (3 nodes → 4 nodes → 2 nodes)
Pass	On topology change DDS msg loss ≤ 5%, mesh re-converge ≤ 8s

S7-2 — Multicast (DDS) Normal Operation

Attribute	Value
Setup	OpenWrt multicast_mode=1, igmp_snooping=1 enabled
Pass	8-robot DDS discovery all success, broadcast msg receive 100%

S7-3 — KPP Latency ≤ 150ms

Attribute	Value
Environment	OpenWrt mesh, 8 robots
Measurement	FollowerTarget (10Hz) publish→receive latency, 100,000 samples
Pass	p99 ≤ 150ms (KPP), p50 ≤ 50ms

4.4 S8 — LTE Failover (mwan3)

S8-1 — Auto LTE Switch on Wi-Fi Loss

Attribute	Value
Environment	Hub UGV SBC #2 mwan3 enabled
Setup	Wi-Fi 5GHz artificially blocked during driving
Pass	LTE backhaul switchover within 8s, operator screen info uninterrupted

S8-2 — Latency in LTE Environment

Attribute	Value
Measurement	LTE backhaul communication latency
Pass	Remote C2 command ≤ 200ms (LTE), operator telemetry ≤ 100ms

5. Test Scenarios — Sprint 9-12 (Safety + Election + Threat)

5.1 S9 — 5-Tier Autonomous Avoidance

S9-1 — T1.5 Auto ±2m Detour (KPP 300ms)

Attribute	Value
Environment	W2 forest, dynamic obstacle during driving
Measurement	Obstacle detect → Tier 1.5 entry → detour start time
Pass	≤ 300ms (KPP), detour success 100%, 0 collisions

S9-2 — Tier 2/3 Catch-Up

Attribute	Value
Setup	Apply artificial 30% speed limit to 1 follower → induce lag
Pass	T2 entry 1.2x speed, T3 entry max speed, recovery transitions T1.5/T1

5.2 S10 — Modified Raft Leader Election

S10-1 — Hub UGV Normal Succession

Attribute	Value
Environment	W1, 8-robot squadron
Setup	Leader Go2 artificially powered off (battery 0%)
Pass	Hub UGV succeeds as priority 1, mission continues

S10-2 — Quorum Unmet Wait

Attribute	Value
Setup	9 → 4 robots reduced then leader lost
Pass	Quorum (5) unmet, 30s wait then retry, "Re-organizing" displayed

S10-3 — KPP Reform ≤ 10s

Attribute	Value
Measurement	Leader OFF → new leader formation stabilize time (100 reps)
Pass	p99 ≤ 10s (KPP), avg ≤ 6s

5.3 S11 — Threat Detection + Fire Control

S11-1 — Drone Detection (RF Scanner)

Attribute	Value
Environment	W5, sim drone spawn at 2km distance
Pass	RF scanner detect ≤ 2km (KPP), DetectedObject publish ≤ 100ms

S11-2 — Fire Authorization Security

Attribute	Value
Setup	Issue fire command without operator PIN
Pass	FireAuthorization rejected, audit log recorded, 0 auto-fires

5.4 S12 — KPP Comprehensive

S12-1 ~ S12-5 — 5 KPP 100 Repetitions (Regression Suite)

Auto-executed in CI. 100 measurements per KPP, p99/p50 distribution analysis.

6. Test Scenarios — Sprint 13-14 (Real Robot)

6.1 S13 — Real Robot 3 Units (Go2 + 2 UGV)

S13-1 — Wi-Fi 6 Mesh Outdoor Verification

Attribute	Value
Location	Internal test site (open 50x50m)
Setup	OpenWrt 24.10 router 3 units, DAWN AP steering, batman-adv
Pass	3-hop mesh operational, throughput ≥ 50 Mbps, latency p99 ≤ 30ms

S13-2 — Real Drone Detection (Optional, Safe Env)

Attribute	Value
Setup	Civilian FPV drone simulated intrusion (with pre-authorization)
Pass	RF scanner detection range, AI classification accuracy ≥ 90%

S13-3 — Pan-Tilt Tracking Precision

Attribute	Value
Setup	500m distance target, 3 m/s laterally moving RC vehicle
Pass	Tracking error ≤ 0.05° (KPP)

6.2 S14 — Real Robot 9 Units (Go2 + 8 UGV)

S14-1 — 8-Robot V Outdoor Drive

Attribute	Value
Location	Integrated training ground (open + partial woodland)
Distance	~500m mission scenario
Pass	Formation hold avg ≤ 2m, comm latency ≤ 150ms

S14-2 — Artificial Leader Loss + Hub Succession

Attribute	Value
Setup	Leader Go2 powered off → Hub UGV

Attribute	Value
	succession
Pass	Reform ≤ 10s, mission continues, operator screen shows "Hub Leader"

S14-3 — 8-Robot Auto Mission (A→F)

Attribute	Value
Setup	"Offensive mode" auto flow — Assembly→Column→V→Assault→Rank→Merge→Column
Pass	All stages auto-transition, assembly success ≥ 95%, avg mission ≤ 10 min

7. Test Scenarios — S15 v1.1 New (6 scenarios)

7.1 S15-1 — 8-Robot 360° Surveillance Distribution Accuracy

Attribute	Value
Scenario ID	S15-1
Environment	Gazebo W5 mixed_combat (sim 8 robots)
Purpose	Verify Leader 10s-period sector assignment accuracy
Duration	10 min

Setup

- 8-robot V-formation (d=5m, θ=60°, Leader S1 + Hub UGV S2 + F1-F6 = S3-S8), W5 environment
- Leader publishes SurveillanceSectorAssignment (10s period)
- Each Follower runs sweep mode (30°/sec) via PanTiltCommand
- Gazebo /world/model ground truth for all robot positions

Test Steps

1. Verify Leader publishes sector_assign every 10s (rosbag2)
2. Each follower performs sweep within assigned sector (Gazebo pan-tilt joints)
3. On threat detection event (sim drone spawn), verify immediate reassignment
4. On 1 follower lost, verify gap-filling reassignment
5. On formation change (V → diamond), verify immediate reassignment

Pass Criteria

- Periodic refresh: 10s ± 0.5s
- 360° coverage: 8 robots (Leader+F1-6+Hub) summed sectors cover 360° (overlap allowed)
- Sweep operating ratio: ≥ 95%
- Threat detection sector focus response: ≤ 1s
- Follower lost gap-filling response: ≤ 2s

7.2 S15-2 — Pan-Tilt Sweep Target Detection Rate

Attribute	Value
Scenario ID	S15-2
Environment	Gazebo W2 forest
Purpose	Verify KPP ≥ 90% target detection during sweep operation while driving

Attribute	Value
Duration	15 min

Setup

- 8-robot squadron W2 driving, Leader 1.0 m/s
- Sweep mode 30°/sec, HFOV 60°
- 50 Gazebo simulated targets (human silhouettes) randomly spawned in sectors
- Measure latency from target appearance to AI detection

Pass Criteria

- Target detection rate ≥ 90% (KPP)
- Detection latency ≤ 2s (exposure within one sweep period)
- False positive < 5%
- Sweep → Track mode transition response < 200ms

7.3 S15-3 — Hub SBC #1/#2 Failure Partial Operation

Attribute	Value
Scenario ID	S15-3
Environment	Gazebo W1 open_field
Purpose	Verify partial operation when one of Hub UGV's dual SBCs fails
Duration	20 min

Test Cases

Case A: SBC #1 (SLAM) Failure

- Setup: Kill Hub UGV SBC #1 process
- Expected: Aggregated SLAM dispatch suspended, each robot continues with local SLAM
- Pass: Comm/video/1s-prediction all normal, operator screen "Global map update suspended" notice

Case B: SBC #2 (Comm) Failure

- Setup: Kill Hub UGV SBC #2 process
- Expected: Other Wi-Fi6 mesh robots assist routing, LTE/video suspended
- Pass: Intra-swarm comm continues, operator screen "Hub comm SBC failure" displayed

Case C: Auto-Reboot Attempt

- Setup: SBC #2 auto-reboot (1 attempt limit)
- Pass: Reboot complete within 60s, video stream recovered

7.4 S15-4 — SLAM 5s Period Global Map Consistency (v1.3 updated)

Attribute	Value
Scenario ID	S15-4
Environment	Gazebo W3 urban_blocks
Purpose	Verify global map consistency KPP at SLAM 30/60s periods
Duration	20 min

Test Cases

Case A: 30s Period (default)

- Setup: SLAMLocalDelta 30s period, 8 robots
- Pass: Post-merge grid alignment error ≤ 0.5m (vs Gazebo ground truth)
- Comm load measured ≤ 5 KB/s (target 1-3 KB/s, 100x reduction verified)

Case B: 60s Period (Wide-Area Recon)

- Pass: Grid alignment error ≤ 0.8m

Case C: 15s Period (Urban Mode)

- Pass: Alignment error ≤ 0.3m, comm load ≤ 10 KB/s

Case D: Immediate Update After Formation Change

- Pass: Immediate (1x) SLAMAggregatedMap broadcast post formation change

7.5 S15-5 — GStreamer SRT 200ms Latency KPP

Attribute	Value
Scenario ID	S15-5
Environment	Real robot (internal test site, LTE environment)
Purpose	Verify GStreamer SRT video streaming KPP "real-time ≤ 200ms"
Duration	30 min (repeat)

Setup

- GStreamer SRT pipeline enabled on Hub UGV SBC #2
- Operator Android APP issues VideoStreamRequest
- LTE backhaul (Hub → internal LTE modem → Operator)
- Measure end-to-end timestamp from test pattern encoding to decoding display

Test Steps

6. Operate FHD 1.5Mbps H.265 SRT live latency=120ms
7. Measure end-to-end latency at Operator APP (10 min, 1Hz)
8. Track latency change on LTE signal strength variation (RSSI -90 to -110 dBm)
9. Switch to HD 3Mbps and repeat measurement
10. Verify SRT ARQ recovery after 5% packet loss injection

Pass Criteria

- E2E latency: p99 ≤ 200ms (KPP), p50 ≤ 150ms
- Frame loss: ≤ 1% (after SRT ARQ recovery)
- Uninterrupted stream despite LTE signal change (auto bitrate)
- AES-128 encryption active (verify passphrase auto-issuance)

7.6 S15-6 — Multi-Follower Simultaneous Video Monitoring

Attribute	Value
Scenario ID	S15-6
Environment	Real robot (Operator APP)
Purpose	Verify simultaneous multi-follower video + auto quality adjustment
Duration	30 min

Test Cases

Case A: 1 Stream HD

- Pass: 1920x1080 30fps 3Mbps stable, latency ≤ 200ms

Case B: 3 Streams Simultaneous (FHD)

- Setup: F1, F3, Hub video simultaneously requested
- Pass: 3 streams 1280x720 30fps 1.5Mbps stable, total 4.5 Mbps

Case C: 4+ Streams → Auto Thumbnail Downgrade

- Setup: 5 follower videos simultaneously requested
- Pass: Auto thumbnail mode (320x240 5fps 100Kbps) entry, total ≤ 500 Kbps

Case D: VideoStreamStop Normal Cleanup

- Setup: Operator taps "Stop View Video"
- Pass: GStreamer pipeline cleanup within 5s, follower CPU usage idle

8. Regression Test (CI) Matrix

8.1 CI Auto-Execution

Regression Scenario	Frequency	Environment	Runtime	Pass Criteria
S1-2, S3-1 (bench)	Per PR ★	bench (dummy)	5 min	FSM transitions OK, message routing OK
S16-1 ~ S16-6 (bench)	Per PR ★	bench (dummy)	5 min	DEMO 6-phase completes normally
S1-2, S2-2	Nightly	Gazebo	10 min	0 collisions, avg error ≤ 2.5m
S4-1, S5-1	Nightly	Gazebo	15 min	10Hz±10%, success ≥ 95%
S7-3	Nightly	Gazebo + netem	30 min	p99 ≤ 150ms
S10-3	Nightly	Gazebo	20 min	p99 ≤ 10s
S15-1, S15-2	Weekly	Gazebo W5/W2	30 min	Sector accuracy, detection ≥ 90%
S15-3, S15-4	Weekly	Gazebo W1/W3	30 min	Partial op + map alignment ≤ 0.5m
Full regression suite	Pre-release	Gazebo + real robot	4 hrs	All KPPs PASS

★ = v1.2 new bench tier (10-min regression with dummy node, no Gazebo)

8.2 Report Format

- rosbag2 (.mcap) — raw message recording
- PlotJuggler analysis (.png + .csv)
- KPP measurement JSON (Jenkins/Grafana visualization)
- Auto Pass/Fail report + Slack alerts

9. S16 — DEMO 6-Phase Sequencer Scenarios (v1.2 new)

Per decision A2, six new scenarios verifying the DEMO 6-phase auto-sequencer. For DAPA demos, VIP tours, and tech verification.

9.1 S16-1 — DEMO 6-Phase Full Flow (bench)

Attribute	Value
Scenario ID	S16-1
Environment	bench (Docker + dummy node)
Purpose	Verify full DEMO IDLE -> FORWARD -> SCAN -> ENGAGE -> REVERSE -> COMPLETE pass
Test time	5 min (PR CI)

Setup

- deployment_mode = bench (yaml overlay applied)
- combat_robot_test_dummy_node active (IMU/Pan-Tilt/Detector/Gun simulated)
- App issues RECON demo command

Pass Criteria

- All 6 phases pass normally, auto transition between phases
- Total sequence completes within 60s (default demo_* distance/duration)
- Final IDLE return + COMPLETE displayed

9.2 S16-2 — DEMO Stop / Resume

Attribute	Value
Scenario ID	S16-2
Environment	bench
Purpose	Verify forced stop (resetDemoSequence) and resume behavior during DEMO

Test Cases

Case A: Stop during SCAN phase

- Setup: Enter SCAN phase, after 5s issue Stop Demo command
- Pass: resetDemoSequence called, immediate IDLE return, actuators stop

Case B: EmergencyStop during ENGAGE phase

- Setup: ENGAGE phase fire started, then EmergencyStop issued

- Pass: Fire halts immediately, EMERGENCY_STOP_STATE entered, gun_trigger_permission=DENIED

Case C: Resume

- Setup: After stop, Restart Demo command
- Pass: Restarts from FORWARD phase

9.3 S16-3 — deployment_mode=demo Weapon Disable Verification

Attribute	Value
Scenario ID	S16-3
Environment	bench + Gazebo
Purpose	Verify ENGAGE phase does not emit PWM fire signal in demo mode

Pass Criteria

- With deployment_mode=demo, gun_trigger_permission=DENIED is forced
- ENGAGE phase: PWM signal stays at 0 (oscilloscope/GPIO measurement)
- Operator screen shows yellow banner "Demo mode — weapons disabled"
- audit log records "deployment_mode=demo, fire suppressed"

9.4 S16-4 — DEMO Multi-Robot Synchronized Demo (Leader + Hub)

Attribute	Value
Scenario ID	S16-4
Environment	Gazebo W1 or real robot (3-robot minimum)
Purpose	Synchronized demo with Leader (S1) + Hub UGV (S2) + 1-2 followers

Setup

- Leader S1, Hub UGV S2, Follower S3 (3-robot minimum)
- Leader issues RECON demo command -> swarm_coordinator relays SwarmRobotCommand to S2, S3
- All robots progress through phases in sync

Pass Criteria

- On phase transition, all 3 robots transit within ±500ms
- FORWARD phase: Hub UGV maintains rear position, Followers maintain V-formation
- SCAN phase: 8-robot sector distribution applies (Leader front, Hub rear, Follower flanks)

9.5 S16-5 — DAPA Demo Regression (Real Robot)

Attribute	Value
Scenario ID	S16-5
Environment	Real robot integrated training ground (full 8 robots)
Purpose	Regression verification for DAPA formal demo scenario
Test time	30 min (3 repetitions)

Pass Criteria

- 8-robot simultaneous demo: phase transition sync within ±1s
- Target detection -> ENGAGE auto-transition -> fire (BB pellet/blank)
- Total demo time within 5 min
- All 3 repetitions PASS

9.6 S16-6 — command_id Echo Verification (A3)

Attribute	Value
Scenario ID	S16-6
Environment	bench (or Gazebo)
Purpose	Verify v1.2 new command_id echo pattern

Test Cases

Case A: FormationCommand command_id echo

- Setup: App issues FormationCommand (command_id=12345)
- Pass: Next RobotStatus message contains last_received_command_id=12345 echo
- All 8 robots echo same command_id (broadcast successful)

Case B: Command miss detection

- Setup: Artificial 1% packet loss
- Pass: App detects last_received_command_id not updated within 5s
- App UI displays "Command not received" alert

Case C: Sequence verification

- Setup: Issue 100 commands consecutively
- Pass: All 100 command_ids echo received 100%, order preserved

10. S17 — Cost Map + LTE Redundancy Scenarios (v1.3 new)

Six new scenarios verifying the Cost Map system (see SDD-SWARM §9.6) and LTE redundancy (§5.5). S17-1~4 are Gazebo simulations, S17-5 is bench (KPP measurement), S17-6 is real robot (LTE failover).

10.1 S17-1 — Robosense E1 Ground Plane Segmentation Accuracy

Attribute	Value
Scenario ID	S17-1
Environment	Gazebo W2 forest (slopes + grass)
Purpose	RANSAC ground segmentation correctly classifies ground points on non-planar terrain
Test time	20 min

Pass Criteria

- Ground point classification accuracy >= 95% (vs ground truth)
- False positive (obstacle as ground) < 2%
- Slope 30 deg threshold accurate (30.5 deg = lethal, 29.5 deg = cautious)

10.2 S17-2 — 235mm Obstacle Detection + Avoidance

Attribute	Value
Scenario ID	S17-2
Environment	Gazebo W1 open_field (obstacle spawn)
Purpose	Verify UGV spec lethal threshold 235mm and avoidance behavior

Test Cases

- Case A: 200mm brick -> detected as inflated (cost 200), slow pass possible
- Case B: 235mm brick -> lethal (cost 254), mandatory detour
- Case C: 270mm brick -> lethal, detour with +/-2m lateral trajectory

Pass Criteria

- Case A: pass success, avg speed >= 1.5 m/s
- Case B/C: 0 collisions, 100% detour success
- Tier 1.5 entry time <= 300 ms (existing KPP)

10.3 S17-3 — 220mm Ditch Detection + Avoidance

Attribute	Value
Scenario ID	S17-3
Environment	Gazebo W4 mountain (ditches/pits)
Purpose	Verify UGV spec ditch threshold 220mm and avoidance

Test Cases

- Case A: 150mm width ditch -> passable (free)
- Case B: 220mm width ditch -> lethal (exact threshold), mandatory detour
- Case C: 300mm deep, >= 220mm width pit -> lethal + alert

Pass Criteria

- Case A: pass success
- Case B/C: detour success, 0 collisions/falls
- LiDAR upward scan provides terrain awareness beyond ditches

10.4 S17-4 — 30 deg Slope Capability Accurate Verification

Attribute	Value
Scenario ID	S17-4
Environment	Gazebo W4 mountain
Purpose	Verify UGV spec slope threshold 30 deg is correctly applied

Test Cases

- Case A: 25 deg slope -> pass (cautious cost 100, slow down)
- Case B: 30 deg slope (threshold) -> pass attempt (climbing limit, 50% speed)
- Case C: 35 deg slope -> lethal, detour

Pass Criteria

- Case A/B: pass success, no slipping
- Case C: lethal classification, detour trajectory generated
- Speed during climbing: at 30 deg <= 1.5 m/s (safe slowdown)

10.5 S17-5 — Cost Map KPP Verification (latency <= 5s, coverage >= 7m)

Attribute	Value
Scenario ID	S17-5

Attribute	Value
Environment	bench (Docker + Gazebo W1, new KPP measurement)
Purpose	v1.3 two new KPPs — cost_map_latency_p99 <= 5s, cost_map_coverage >= 7m

Setup

- UGV driving at max speed 12 km/h (3.33 m/s)
- 100 repetitions, p99 latency computed
- Measure /local/cost_map publish time - sensor input time difference

Pass Criteria

- cost_map_latency_p99 <= 5 sec (measurement target: < 1 sec, 5x margin)
- cost_map_coverage forward >= 7m (14m area published at all times)
- p99 update rate 0.8~1.2 Hz (normal)
- 3 consecutive >5s -> Node Watchdog ERROR_STATE transition confirmed

10.6 S17-6 — LTE Redundancy Fail-over (Real Robot)

Attribute	Value
Scenario ID	S17-6
Environment	Real robot (Hub S2 + S3 LTE backup + operator)
Purpose	§5.5 LTE redundancy — When Hub LTE fails, Follower S3 auto-backup

Test Cases

- Case A: Hub LTE modem normal -> S3 in LTE_BACKUP_DESIGNATED (standby)
- Case B: Hub LTE modem artificially disabled -> within 8s S3 LTE_PROMOTED, lte_term += 1
- Case C: After S3 LTE active, verify operator <-> S3 <-> swarm communication
- Case D: Hub LTE recovery -> Hub PROMOTED, S3 DEMOTED, lte_term += 1

Pass Criteria

- Case B: backup activation time <= 10s (8s mwan3 detect + 2s activation)
- Case C: Operator telemetry uninterrupted (<= 30s temporal gap)
- Case D: No split-brain (only 1 LTE_PROMOTED at any time)
- LTERoleAnnouncement broadcast received normally (all 8 robots)

11. Deputy + Limp Mode Scenarios (S18-1 ~ S18-6, ★ v1.5 new — DCN-2026-001 D-006)

Six new scenarios verifying the Hub-Deputy redundancy and 4-tier Leader succession policies defined in SAN-SDD-SWARM-001 v1.5 §5.5 / §5.6 / §5.7. All six are automated by san_l5_regression scenario_runner.cpp and run as part of the L1 bench regression on every PR. Each scenario uses a uniform Setup / Pass Criteria template; deadlines are enforced by corresponding parameters in scenario_runner.hpp (s18_1_deadline_ms .. s18_6_deadline_ms).

11.1 S18-1 — Leader → Deputy Succession (1st Priority)

Attribute	Value
Scenario ID	S18-1
Environment	bench (Docker dummy injection) + Gazebo W1
Purpose	Verify that when Leader (S1) fails, Deputy UGV (S3) auto-promotes as 1st-priority Leader within ≤ 5 s.
Pre-condition	8-robot baseline seeded (S1 active leader; S3 deputy with SBC1+SBC2 healthy, battery ≥ 20%)
Deadline	≤ 5 s (5000 ms)
Implementing test	san_role_management/test/test_leader_succession.cpp, san_l5_regression::runS18_1_LeaderToDeputy()

11.1.1 Setup

- seedEightRobotBaseline(): S1~S8 all sbc1_healthy=true; S2/S3 sbc2_healthy=true; S3 is_deputy_ugv=true; S1 is_leader_role_active=true. Battery 79..72%
- injector_->removeRobot(1): S1 RobotStatus publish stops → LeaderRoleManager watchdog times out after 1.4 s

11.1.2 Pass Criteria

- Receive on /swarm/leader/role_announce within ≤ 5 s with: role == LEADER_PROMOTED
- robot_id == 3 (Deputy UGV)
- succession_priority == PRIORITY_DEPUTY (= 1)
- leader_term incremented by ≥ 1 from previous value
- No other robot publishes PROMOTED in the same window (split-brain absent)

11.2 S18-2 — Leader + Deputy both fail → Hub Succession (2nd Priority)

Attribute	Value
Scenario ID	S18-2

Attribute	Value
Environment	bench + Gazebo W1
Purpose	Verify that when both Leader (S1) and Deputy (S3) fail, Hub UGV (S2) auto-promotes as 2nd-priority Leader within ≤ 8 s.
Pre-condition	8-robot baseline seeded
Deadline	≤ 8 s (8000 ms)
Implementing test	san_l5_regression::runS18_2_LeaderDeputyToHub()

11.2.1 Setup

- Baseline seeded
- removeRobot(1) + removeRobot(3): S1 (Leader) and S3 (Deputy) stopped simultaneously
- S2 (Hub) sbc1/sbc2 both healthy, battery ≥ 20%

11.2.2 Pass Criteria

- Within ≤ 8 s receive LeaderRoleAnnouncement(role=LEADER_PROMOTED, robot_id=2, succession_priority=HUB)
- leader_term incremented
- 8 s budget breakdown: Deputy grace 200 ms + Hub grace 400 ms + Deputy timeout 5 s + safety margin

11.3 S18-3 — Hub fails → Deputy Full Takeover (LTE + SLAM + Video)

Attribute	Value
Scenario ID	S18-3
Environment	bench + Gazebo W1 + real-robot Phase 2
Purpose	Verify that when Hub UGV (S2) fails, Deputy (S3) takes over LTE gateway + SLAM aggregation + GStreamer SRT relay ALL THREE within ≤ 7 s (partial takeover not allowed).
Pre-condition	8-robot baseline; Leader (S1) remains healthy (no Leader succession in this scenario)
Deadline	≤ 7 s (7000 ms)
Implementing test	san_role_management/test/test_hub_deputy_takeover.cpp, san_l5_regression::runS18_3_HubToDeputyTakeover()

11.3.1 Setup

- Baseline seeded
- removeRobot(2): Hub UGV total failure (Leader unaffected)

11.3.2 Pass Criteria

- Within ≤ 7 s receive HubRoleAnnouncement(role=HUB_PROMOTED, robot_id=3)
- ALL three capability flags simultaneously true:
-   - lte_active == true
-   - slam_aggregation_active == true
-   - video_relay_active == true
- hub_term incremented by ≥ 1
- LTE-only or SLAM-only partial takeover → FAIL (covered separately by §5.14 LTERoleAnnouncement tests)

11.4 S18-4 — Leader + Hub + Deputy all fail → Battery-max Follower (3rd Priority)

Attribute	Value
Scenario ID	S18-4
Environment	bench + Gazebo W1
Purpose	Verify that when all three priority-1/2 candidates fail, the battery-max follower auto-promotes as 3rd-priority Leader within ≤ 10 s.
Pre-condition	8-robot baseline; battery deliberately distributed S4=60%, S5=90%, S6=45%, S7=55%, S8=30% (S5 is unique winner)
Deadline	≤ 10 s (10000 ms)
Implementing test	san_role_management/test/test_battery_selection.cpp, san_l5_regression::runS18_4_ThreeFailedBatteryFollower()

11.4.1 Setup

- Baseline seeded with battery distribution above
- removeRobot(1) + removeRobot(2) + removeRobot(3): Leader + Hub + Deputy all stopped

11.4.2 Pass Criteria

- Within ≤ 10 s receive LeaderRoleAnnouncement(role=LEADER_PROMOTED, succession_priority=BATTERY_MAX)
- Promoted robot_id == 5 (S5, battery 90%, unique max)
- Tie-breaker rule (equal battery → lowest robot_id wins) covered by separate test
- 10 s budget breakdown: Deputy/Hub grace + 5 s timeout + 200 ms BATTERY_MAX grace + accumulation

11.5 S18-5 — Hub + Deputy both fail → Limp Mode Entry

Attribute	Value
Scenario ID	S18-5

Attribute	Value
Environment	bench + Gazebo W1
Purpose	Verify that when both Hub (S2) and Deputy (S3) fail simultaneously, Limp Mode entry alert is published within ≤ 8 s. (Leader S1 remains healthy.)
Pre-condition	8-robot baseline; Leader (S1) healthy
Deadline	≤ 8 s (8000 ms)
Implementing test	san_role_management/test/test_limp_mode.cpp, san_l5_regression::runS18_5_HubDeputyBothDownLimpEnter()

11.5.1 Setup

- Baseline seeded
- removeRobot(2) + removeRobot(3): S2 (Hub) and S3 (Deputy) stopped simultaneously

11.5.2 Pass Criteria

- Within ≤ 8 s receive message on /swarm/limp_mode/alert with data containing "LIMP_MODE_ACTIVE"
- Additional verification (covered by integration tests):
- - Wi-Fi 6 mesh routing reconverges excluding Hub/Deputy nodes
- - Android app switches to mesh-direct mode via nearest surviving Follower
- - Fire/strike command interface remains ENABLED (D-004 Option A verification)
- - GStreamer video sends directly from each robot to Android IP (no Hub relay)

11.6 S18-6 — Deputy Recovers → Limp Mode Exit

Attribute	Value
Scenario ID	S18-6
Environment	bench + Gazebo W1
Purpose	Verify that after Limp Mode is entered (S18-5), Deputy (S3) recovery triggers Limp Mode exit alert + Hub-role takeover within ≤ 5 s.
Pre-condition	System in Limp Mode (S2 + S3 both down, Leader S1 still healthy)
Deadline	≤ 5 s (5000 ms)
Implementing test	san_l5_regression::runS18_6_DeputyRecoversLimpExit()

11.6.1 Setup

- Enter Limp Mode per S18-5 (S2 + S3 down) and verify alert received
- Reset alert watcher

- restoreRobot(3, {sbc1_healthy=true, sbc2_healthy=true, battery_percent=75.0, is_deputy_ugv=true}): Deputy comes back online

11.6.2 Pass Criteria

- Within ≤ 5 s receive /swarm/limp_mode/alert containing "LIMP_MODE_EXITED"
- Within ≤ 5 s also receive HubRoleAnnouncement(role=HUB_PROMOTED, robot_id=3) with all three flags true
- Hub (S2) recovery in a subsequent step would trigger HUB_DEMOTED (covered by S18-3 reverse test)

11.7 S18 Scenario Summary Matrix

Scenario	Trigger	Expected Succession	Deadline	Implementing Function
S18-1	Leader (S1) down	Deputy (S3) → Leader	≤ 5 s	runS18_1_LeaderToDeputy()
S18-2	Leader + Deputy down	Hub (S2) → Leader	≤ 8 s	runS18_2_LeaderDeputyToHub()
S18-3	Hub (S2) down	Deputy (S3) → Hub (full)	≤ 7 s	runS18_3_HubToDeputyTakeover()
S18-4	Leader + Hub + Deputy down	Battery-max follower (S5) → Leader	≤ 10 s	runS18_4_ThreeFailedBatteryFollower()
S18-5	Hub + Deputy down	LIMP_MODE_ACTIVE alert	≤ 8 s	runS18_5_HubDeputyBothDownLimpEnter()
S18-6	Deputy recovers from Limp	LIMP_MODE_EXITED + Deputy → Hub	≤ 5 s	runS18_6_DeputyRecoversLimpExit()

12. Scenario Index (66 total)

11.1 Simulation + bench (54)

#	ID	Theme	Env	KPP
1	S1-1	Stationary leader tracking	W1	—
2	S1-2	Linear V tracking	W1	form ≤2m
3	S1-3	Curved V tracking	W1	form ≤2m
4	S2-1	Hybrid A* static	W2	—
5	S2-2	Hybrid A* dynamic	W2	form ≤2m
6	S3-1	10Hz prediction broadcast	W1	comm ≤150ms
7	S4-1	V→column transition	W1	assemble ≥95%
8	S4-2	Column→diamond	W1	assemble ≥95%
9-13	S5-1~5	9 formations verification	W1, W4	—
14-15	S6-1~2	9-robot V stability	W1	form ≤2m
16-18	S7-1~3	OpenWrt Mesh + DDS	Sim	comm ≤150ms
19-20	S8-1~2	LTE Failover	Sim	—
21-22	S9-1~2	5-Tier autonomous avoidance	W2	react ≤300ms
23-25	S10-1~3	Modified Raft Election	W1	reform ≤10s
26-27	S11-1~2	Threat detect + fire control	W5	detect ≤2km
28-32	S12-1~5	KPP comprehensive (regression)	W1	5 KPPs
33-35	S13-1~3	Real robot 3	Internal	—

#	ID	Theme	Env	KPP
36-38	S14-1~3	Real robot 9	Range	5 KPPs
39	S15-1	* 360° surveillance	W5	—
40	S15-2	* Pan-tilt sweep detect	W2	classify ≥90%
41	S15-3	* Hub dual SBC partial op	W1	—
42	S15-4	* SLAM 30/60s consistency	W3	—
43	S16-1	** DEMO 6-Phase full flow	bench	—
44	S16-2	** DEMO Stop/Resume	bench	—
45	S16-3	** DEMO Weapon disable verify	bench+Gazebo	—
46	S16-4	** DEMO Multi-robot sync	Gazebo	—
47	S16-6	** command_id Echo	bench	—
48	S17-1	*** E1 ground segmentation	W2	—
49	S17-2	*** 235mm obstacle detect/avoid	W1	react <=300ms
50	S17-3	*** 220mm ditch detect/avoid	W4	—
51	S17-4	*** 30 deg slope verification	W4	—
52	S17-5	*** Cost Map KPP	bench	lat <=5s, cov >=7m
53	S15-4	* SLAM 5s consistency (updated)	W3	—

11.2 Real Robot (6)

#	ID	Theme	Env	KPP
43	S13-1	Wi-Fi6 mesh outdoor	Internal site	comm ≤150ms
44	S13-2	Real drone detect	Internal site (optional)	detect ≤2km
45	S13-3	Pan-tilt track	Internal site	track ≤0.05°
46	S14-1	9-robot V outdoor	Range	form ≤2m
47	S14-2	Leader loss + succession	Range	reform ≤10s
48	S15-5	* GStreamer SRT 200ms	Internal LTE	video ≤200ms
49	S16-5	** DAPA Demo Regression	Training ground (full 8)	—
50	S17-6	*** LTE Redundancy fail-over	Real robot (Hub + S3)	<=10s backup activation

* = v1.1 new, ** = v1.2 new (DEMO 6-phase), *** = v1.3 new (Cost Map + LTE)